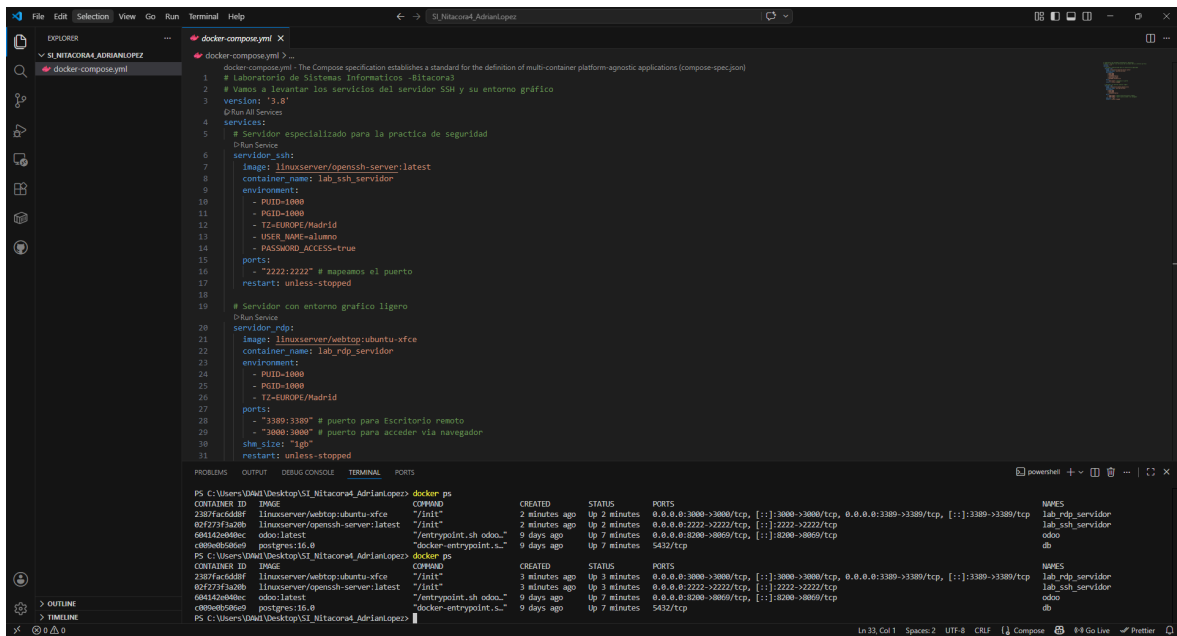


## Bitácora Técnica IV



Primero he copiado todo los datos del servidor y los he subido con el comando de *docker-compose up -d* y comprobado con *docker ps* que están bien los contenedores.



```
1 # Laboratorio de Sistemas Informaticos - Bitacora3
2 # Vamos a levantar los servicios del servidor SSH y su entorno gráfico
3 version: '3.8'
4
5 # Servidor especializado para la practica de seguridad
6 D-Run Service
7
8 servidor_ssh:
9   image: linuxserver/openssh-server:latest
10   container_name: lab_ssh_servidor
11   environment:
12     - PUID=1000
13     - PGID=1000
14     - TZ=EUROPE/Madrid
15     - USER_NAME=alumno
16     - PASSWORD_ACCESS=true
17   ports:
18     - "2222:2222" # mapeamos el puerto
19   restart: unless-stopped
20
21 # Servidor con entorno gráfico ligero
22 D-Run Service
23
24 servidor_rdp:
25   image: linuxserver/webtop:ubuntu-xfce
26   container_name: lab_rdp_servidor
27   environment:
28     - PUID=1000
29     - PGID=1000
30     - TZ=EUROPE/Madrid
31   ports:
32     - "3380:3380" # puerto para Escritorio remoto
33     - "3800:3800" # puerto para acceder via navegador
34   shm_size: "1gb"
35   restart: unless-stopped
```

| CONTAINER ID | IMAGE                             | COMMAND                  | CREATED       | STATUS       | PORTS                                                                                    | NAMES            |
|--------------|-----------------------------------|--------------------------|---------------|--------------|------------------------------------------------------------------------------------------|------------------|
| 23b7fa6dd8f  | linuxserver/webtop:ubuntu-xfce    | "/init"                  | 2 minutes ago | Up 2 minutes | 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp, 0.0.0.0:3380->3380/tcp, [::]:3380->3380/tcp | lab_rdp_servidor |
| 6027273a2db  | linuxserver/openssh-server:latest | "/init"                  | 2 minutes ago | Up 2 minutes | 0.0.0.0:2222->2222/tcp, [::]:2222->2222/tcp                                              | lab_ssh_servidor |
| 6041c4e00ec  | odoo:latest                       | "/entrypoint.sh odoo"    | 9 days ago    | Up 9 minutes | 0.0.0.0:8200->8080/tcp, [::]:8200->8080/tcp                                              | odoo             |
| c080e0500e9  | postgres:16.0                     | "docker-entrypoint.s..." | 9 days ago    | Up 7 minutes | 5432/tcp                                                                                 | db               |

| CONTAINER ID | IMAGE                             | COMMAND                  | CREATED       | STATUS       | PORTS                                                                                    | NAMES            |
|--------------|-----------------------------------|--------------------------|---------------|--------------|------------------------------------------------------------------------------------------|------------------|
| 23b7fa6dd8f  | linuxserver/webtop:ubuntu-xfce    | "/init"                  | 3 minutes ago | Up 3 minutes | 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp, 0.0.0.0:3380->3380/tcp, [::]:3380->3380/tcp | lab_rdp_servidor |
| 6027273a2db  | linuxserver/openssh-server:latest | "/init"                  | 3 minutes ago | Up 3 minutes | 0.0.0.0:2222->2222/tcp, [::]:2222->2222/tcp                                              | lab_ssh_servidor |
| 6041c4e00ec  | odoo:latest                       | "/entrypoint.sh odoo"    | 9 days ago    | Up 7 minutes | 0.0.0.0:8200->8080/tcp, [::]:8200->8080/tcp                                              | odoo             |
| c080e0500e9  | postgres:16.0                     | "docker-entrypoint.s..." | 9 days ago    | Up 7 minutes | 5432/tcp                                                                                 | db               |

```
AdrianLopezAtalaya_Bitacora4
docker-compose.yml X
docker-compose.yml
1 # Laboratorio de Sistemas Informaticos -Bitacora3
2 # Vamos a levantar los servicios del servidor SSH y su entorno gráfico
3 version: '3.8'
4 Run All Services
5 services:
6 # Servidor especializado para la practica de seguridad
7 Run Service
8 servidor_ssh:
9 image: linuxserver/openssh-server:latest
10 container_name: lab_ssh_servidor
11 environment:
12 - PUID=1000
13 - PGID=1000
14 - TZ=EUROPE/Madrid
15 - USER_NAME=alumno
16 - USER_PASSWORD=sistemas_informaticos
17 - PASSWORD_ACCESS=true
18 ports:
19 - "2222:2222" # mapeamos el puerto
20 restart: unless-stopped
21 # Servidor con entorno grafico ligero
22 Run Service
23 servidor_rdp:
24 image: linuxserver/webtop:ubuntu-xfce
25 container_name: lab_rdp_servidor
26 environment:
27 - PUID=1000
28 - PGID=1000
29 - TZ=EUROPE/Madrid
30 - USER_NAME=alumno
31 - USER_PASSWORD=sistemas_informaticos
32 - PASSWORD_ACCESS=true
33 ports:
34 - "3389:3389" # mapeamos el puerto
35 restart: unless-stopped

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
.0.0:3000->3000/tcp, [::]:3000->3000/tcp, 0.0.0.0:3389->3389/tcp, [::]:3389->3389/tcp lab
_rdp_servidor
b1dcba315c07 linuxserver/openssh-server:latest "/init" 2 hours ago Up 2 hours 0.0
.0.0:2222->2222/tcp, [::]:2222->2222/tcp lab
_ssh_servidor
PS C:\Users\Adrián 21\Desktop\AdriánLópezAtalaya_Bitacora4> ssh alumno@localhost -p 2222
The authenticity of host '[localhost]:2222 ([::1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:xdh2Ale2A6h+dgpha4/OoYggaFyNAI6tVYyB7PkaN44.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? no
Host key verification failed.
PS C:\Users\Adrián 21\Desktop\AdriánLópezAtalaya_Bitacora4> ssh alumno@127.0.0.1 -p 2222
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:xdh2Ale2A6h+dgpha4/OoYggaFyNAI6tVYyB7PkaN44.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
alumno@127.0.0.1's password:
Welcome to OpenSSH Server
b1dcba315c07:~$
```

Una vez lo pruebo en mi casa después de los mil problemas para conectarme de clase, se conecta esta vez a la primera al usar el código `ssh alumno@127.0.0.1`



```
19
20 # Servidor con entorno grafico ligero
21 >Run Service
22 servidor_rdp:
23   image: linuxserver/webtop:ubuntu-xfce
24   container_name: lab_rdp_servidor
25   environment:
26     - AUTO_1000

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ssh + - [ ] [ ] ... [ ] [ ] X
.0.0:3000->3000/tcp, [::]:3000->3000/tcp, 0.0.0.0:3389->3389/tcp, [::]:3389->3389/tcp la
_rdp_servidor
b1dcba315c07 linuxserver/openssh-server:latest "/init" 2 hours ago Up 2 hours 0.
.0.0:2222->2222/tcp, [::]:2222->2222/tcp la
_ssh_servidor
PS C:\Users\Adrián 21\Desktop\AdriánLópezAtalaya_Bitacora4> ssh alunnc@localhost -p 2222
The authenticity of host '[localhost]:2222 ([::1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:xdh2Ale2A6h+dgpha4/OoYggaFyNAI6tVYyB7PkaN44.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? no
Host key verification failed.
PS C:\Users\Adrián 21\Desktop\AdriánLópezAtalaya_Bitacora4> ssh alunnc@127.0.0.1 -p 2222
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:xdh2Ale2A6h+dgpha4/OoYggaFyNAI6tVYyB7PkaN44.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
alunnc@127.0.0.1's password:
Welcome to OpenSSH Server
b1dcba315c07:~$ ssh-keygen -t ed25519 -C "adrianlopez.25@campuscamara.es"
```

Ahora ya he añadido el código para la key

```
22   image: linuxserver/webtop:ubuntu-xfce
23   container_name: lab_rdp_servidor
24   environment:
25     - AUTO_1000

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ssh + - [ ] [ ] ... [ ] [ ] X
b1dcba315c07:~$ ssh-keygen -t ed25519 -C "adrianlopez.25@campuscamara.es"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/config/.ssh/id_ed25519):
Enter passphrase for "/config/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /config/.ssh/id_ed25519
Your public key has been saved in /config/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Nw8VuCX5382qKxQAVowIzpE5vUSaTtNAf9pqVIjflZQ adrianlopez.25@campuscamara.es
The key's randomart image is:
+--[ED25519 256]--+
| .+*.0=0. 0. |
| o+0+0..E.+ .. |
| 0o+.0 0. =. |
| o o.* . 0.. |
| . + o S +. . 0. |
| . . ..+ . + |
| o . . . |
| . . . |
| .oo |
+-----[SHA256]-----+
b1dcba315c07:~$
```



Después de dar un par de enters finalmente accede de forma automática.

```
19
20 # Servidor con entorno grafico ligero
  >Run Service
21 servidor_rdp:
22   image: linuxserver/webtop:ubuntu-xfce
23   container_name: lab_rdp_servidor
24   environment:
25     PORT: 1000
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ssh + - [ ] [ ] ... [ ]

```
|          .oo |
+----[SHA256]-----+
b1dcba315c07:~$ ssh-copy-id -p 2222 alumnc@localhost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/config/.ssh/id_ed25519.pub"
The authenticity of host '[localhost]:2222 ([::1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:xdh2Ale2A6h+dgpha4/OoYggaFyNAI6tVYyB7PkaN44
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any
t are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it
s to install the new keys
alumnc@localhost's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -p 2222 'alumnc@localhost'"
and check to make sure that only the key(s) you wanted were added.

b1dcba315c07:~$
```

Ahora he transferido la llave al servidor.

```
23   container_name: lab_rdp_servidor
24   environment:
25     PORT: 1000
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ssh + - [ ] [ ] ... [ ]

```
b1dcba315c07:~$ ssh-copy-id -p 2222 alumnc@localhost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/config/.ssh/id_ed25519.pub"
The authenticity of host '[localhost]:2222 ([::1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:xdh2Ale2A6h+dgpha4/OoYggaFyNAI6tVYyB7PkaN44
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any tha
t are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it i
s to install the new keys
alumnc@localhost's password:

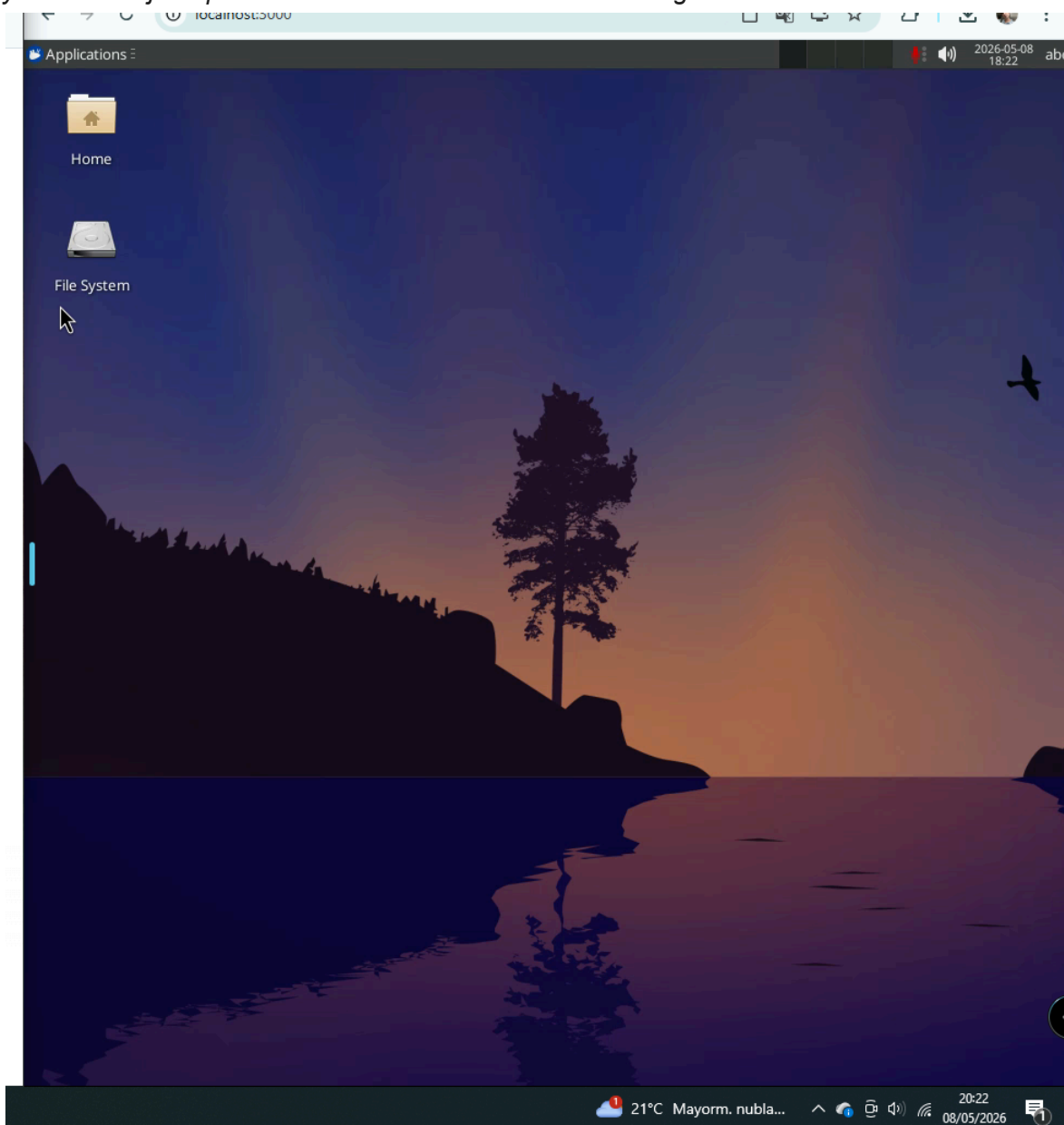
Number of key(s) added: 1

Now try logging into the machine, with: "ssh -p 2222 'alumnc@localhost'"
and check to make sure that only the key(s) you wanted were added.

b1dcba315c07:~$ ssh alumnc@localhost -p 2222
Welcome to OpenSSH Server
b1dcba315c07:~$
```



*ya me ha dejado poder entrar desde el usuario localhost gracias a la clave.*

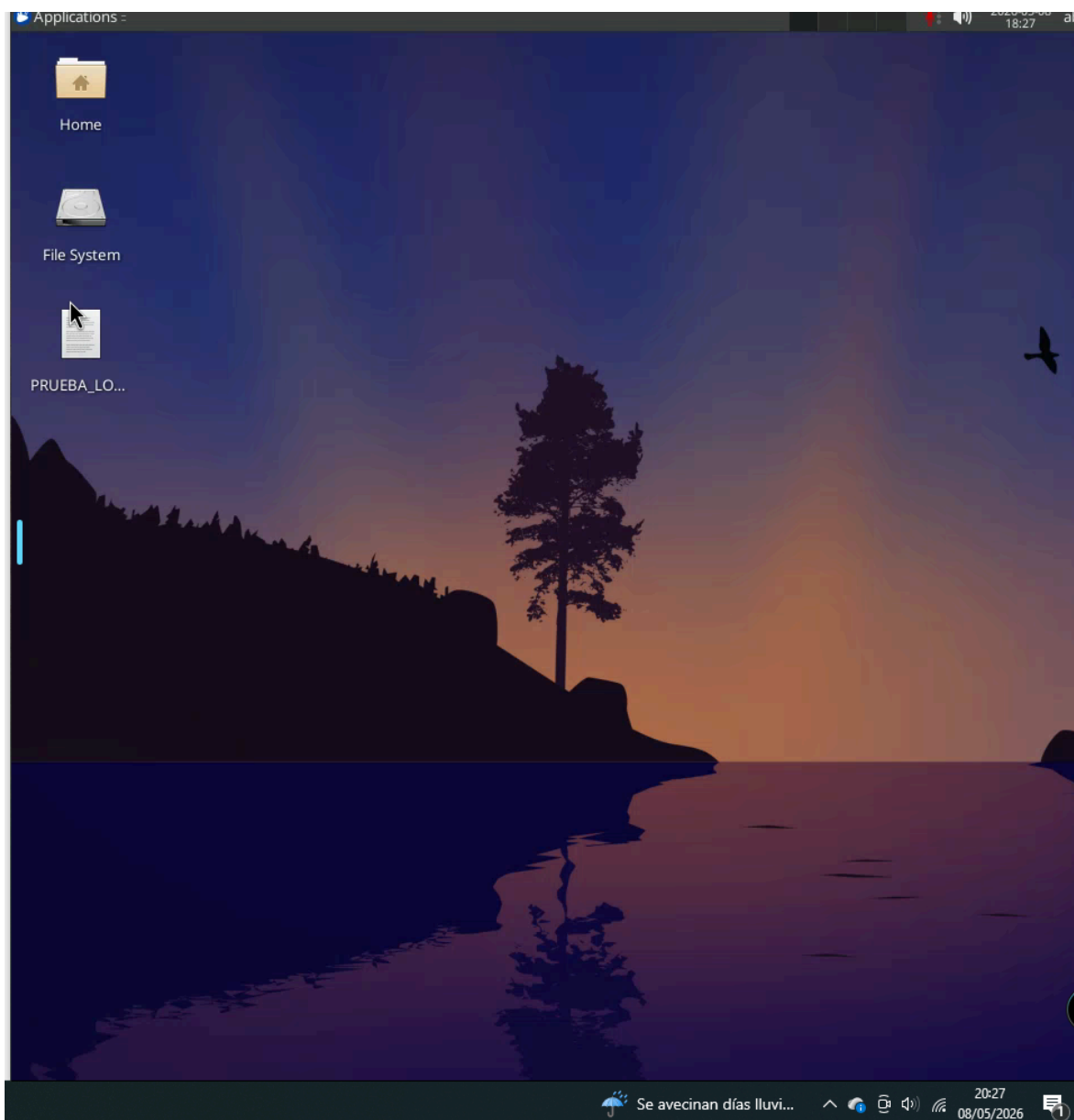


*Ahora accedo mediante el localhost3000 y paso a crear el txt especificado con sus comentarios dentro.*



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*Finalmente considero que ssh es un pilar clave sobre todo en eficiencia, seguridad y automatización.*

*Respecto a la eficiencia destaca que es un protocolo basado puramente en texto. Mientras que por ejemplo RDP necesita una interfaz gráfica completa para poder funcionar (SSH solo transmite caracteres).*

*Pasando a la seguridad, Un servidor gráfico (RPD) es un software complejo con millones de líneas de código, lo que significa más vulnerabilidades potenciales. SSH al ser ligero, robusto y permitir una gestión de identidades muy estricta (como he podido observar en clase a la hora de intentar conectarme y no dejarme en ninguna ocasión) mediante criptografía de clave pública, eliminando el riesgo de ataques contra contraseñas.*



*En resumen, para un informático, el rendimiento es lo primordial, el escritorio remoto es un lujo costoso e innecesario cuando podemos controlar todo el sistema mediante una terminal segura.*

